

PAPER_1306 — Majorana vs Dirac Neutrino Nature

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** C — Particle Physics Open (CLOSED) **Date:** June 11, 2026 **Calculator surface:** `calculate_paradox({"paradox": "majorana_vs_dirac"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1306})`

Master Expression

$F_{\text{TRZ}} \neq 0 \rightarrow$ Majorana mass term permitted via lepton number violation

Match: STRUCTURAL

Interpretation

$F_U = 1$ closure admits Majorana terms when F_{TRZ} break TRZ symmetry. Testable via $0\nu\beta\beta$.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $SO_5 = 10$, $A_5 = 60$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$
- $\Lambda_{\text{QCD}} = 0.217 \text{ GeV}$, $v_{\text{Higgs}} = 246 \text{ GeV}$ (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.

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